Lista 8: Introduction to Computational Neuroscience

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Models and numerical methods

All simulations use three single-compartment Hodgkin–Huxley-type neurons: an excitatory pyramidal cell (E, Olufsen et al.), a fast-spiking basket interneuron (I, Wang–Buzsáki) and an oriens-lacunosum-moleculare interneuron (O, Tort et al.), with the equations given in the problem statement. The synaptic current is

$$J_{\mathrm{sin},ij}(t) = \bar{g}_{ij}\, s_i(t)\,\big(E_{\mathrm{rev}} - V_j(t)\big), \qquad \frac{ds}{dt} = \frac{1+\tanh(V_{\mathrm{pre}}/4)}{2}\,\frac{1-s}{\tau_s} - \frac{s}{\tau_d},$$

where i is the presynaptic and j the postsynaptic neuron. The sign of J_{sin} is set by the driving force $E_{\mathrm{rev}} - V_j$: AMPA synapses ($E_{\mathrm{rev}} = 0\,\mathrm{mV}$) depolarize and $\mathrm{GABA_A}$ synapses ($E_{\mathrm{rev}} = -80\,\mathrm{mV}$) hyperpolarize the postsynaptic cell, because V_j always lies between these values.

The equations were integrated with the fourth-order Runge–Kutta method and $\Delta t = 0.01\,\mathrm{ms}$. A spike is registered at each upward crossing of $0\,\mathrm{mV}$, and its time is obtained by linear interpolation inside the time step. The Fortran program, the shell script that runs all cases and the Python plotting script are listed in the appendices. Random initial voltages (uniform in $[-70, -60]\,\mathrm{mV}$) use a fixed seed, so every number below is reproducible; the effect of other seeds was not tested. As a check on the O-neuron equations, the initial values prescribed in the statement ($m = 0.0122$, $n = 0.0756$, $a = 0.0229$ at $V = -75.61\,\mathrm{mV}$) coincide with the stationary values m_∞, n_∞, a_∞ computed from Eqs. (15), (17) and (18).

Question 1: Synaptic interaction between two neurons

This question investigates how a presynaptic neuron affects a postsynaptic neuron through a unidirectional synapse. In all items the neurons start at $V(0) = -65\,\mathrm{mV}$ with h and n at their stationary values, and the external current is applied only to the presynaptic neuron. For (a) and (b) the pulse lasts from $t = 5\,\mathrm{ms}$ to $t = 15\,\mathrm{ms}$.

(a) Excitatory-to-inhibitory connection

The presynaptic neuron is E, the postsynaptic neuron is I, and $\bar{g}_{EI} = 0.05\,\mathrm{mS/cm^2}$. The simulation lasts 30 ms. Since the statement only asks for *a* value of J_0 giving a single presynaptic spike, a scan of J_0 was performed (file `data/q1_varreduras.txt`): $J_0 = 0.2$ and $0.4\,\mu\mathrm{A/cm^2}$ produce no spike, every tested value from 0.6 to $3.8\,\mu\mathrm{A/cm^2}$ produces

exactly one E spike, and $J_0 = 4$ and $5\,\mu\text{A/cm}^2$ produce two. The value $J_0 = 3\,\mu\text{A/cm}^2$ is used in Figure 1.

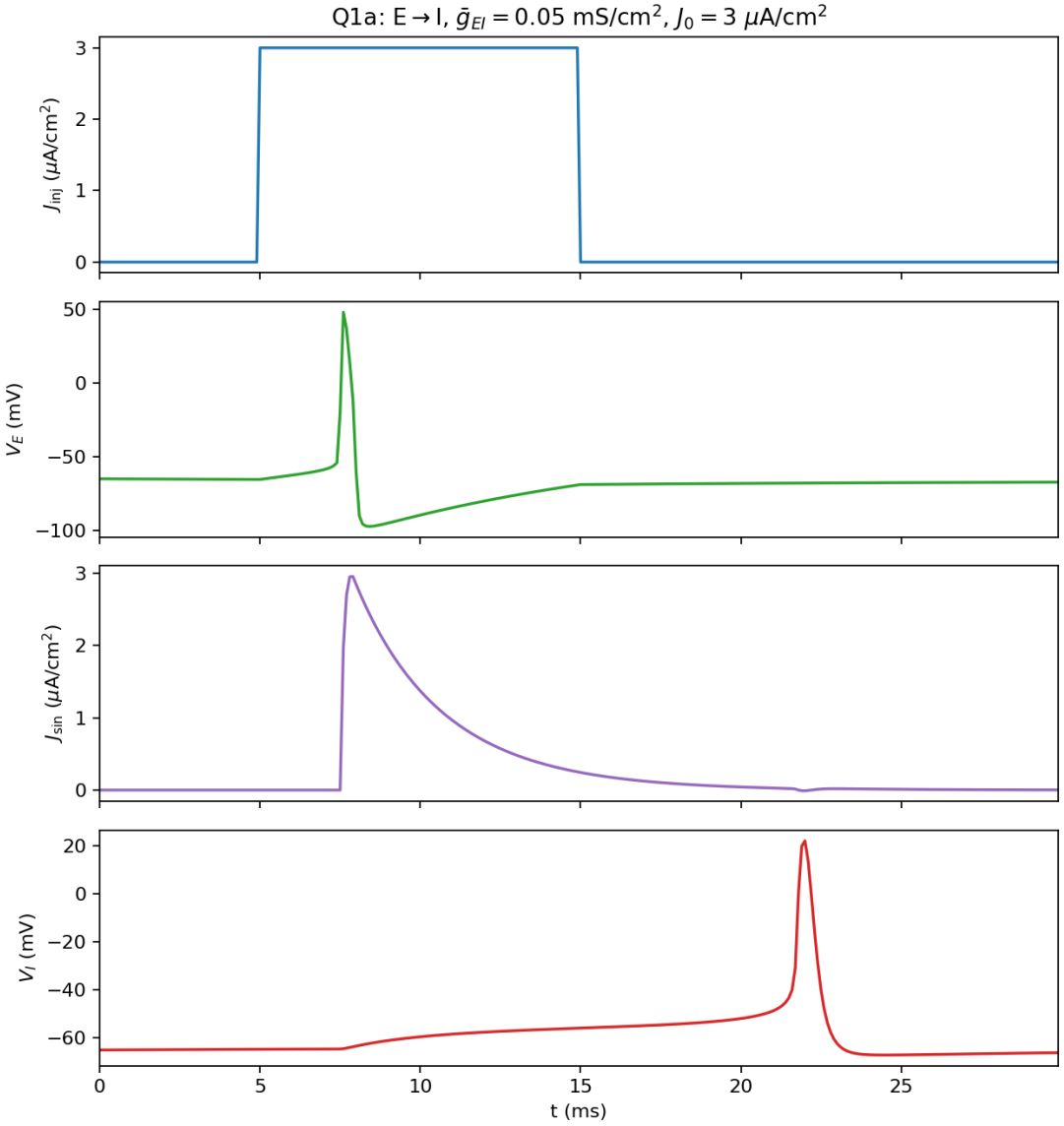


Figure 1: Question 1(a): E→I connection with $\bar{g}_{EI} = 0.05\,\text{mS/cm}^2$ and $J_0 = 3\,\mu\text{A/cm}^2$. From top to bottom: applied current, presynaptic voltage V_E, synaptic current J_{sin}, and postsynaptic voltage V_I.

The E neuron emits one spike at $t = 7.5\,\text{ms}$. The synaptic variable s rises with the fast time constant $\tau_s = 0.1\,\text{ms}$ (peak $s \simeq 0.93$) and decays with $\tau_d = 3\,\text{ms}$, so J_{sin} is a brief, positive (depolarizing) pulse with peak $\simeq 2.95\,\mu\text{A/cm}^2$. The I neuron responds with one spike at $t = 21.8\,\text{ms}$, i.e. 14.3 ms after the presynaptic spike. The postsynaptic spike is therefore not immediate: the brief AMPA current only brings V_I to threshold after a latency, and the long latency is consistent with the Wang–Buzsáki cell being an excitable neuron that can fire at arbitrarily low rates near threshold. After the spike V_I undershoots to $-67.1\,\text{mV}$ and relaxes back towards rest.

(b) Inhibitory-to-excitatory connection

The roles are exchanged: I is presynaptic and receives the pulse, E is postsynaptic, and $\bar{g}_{IE} = 0.1\,\mathrm{mS/cm^2}$. The simulation was extended to 100 ms. The scan shows that the I neuron needs a different current than the E neuron: $J_0 = 0.5\,\mu\mathrm{A/cm^2}$ gives no spike, $J_0 = 1$, 1.5 and $2\,\mu\mathrm{A/cm^2}$ give exactly one, and $J_0 = 3$ and $4\,\mu\mathrm{A/cm^2}$ give two. A current of $4\,\mu\mathrm{A/cm^2}$ therefore does *not* satisfy the one-spike requirement (two I spikes occur, at 8.6 ms and 14.8 ms), whereas $J_0 = 2\,\mu\mathrm{A/cm^2}$ does and is used in Figure 2.

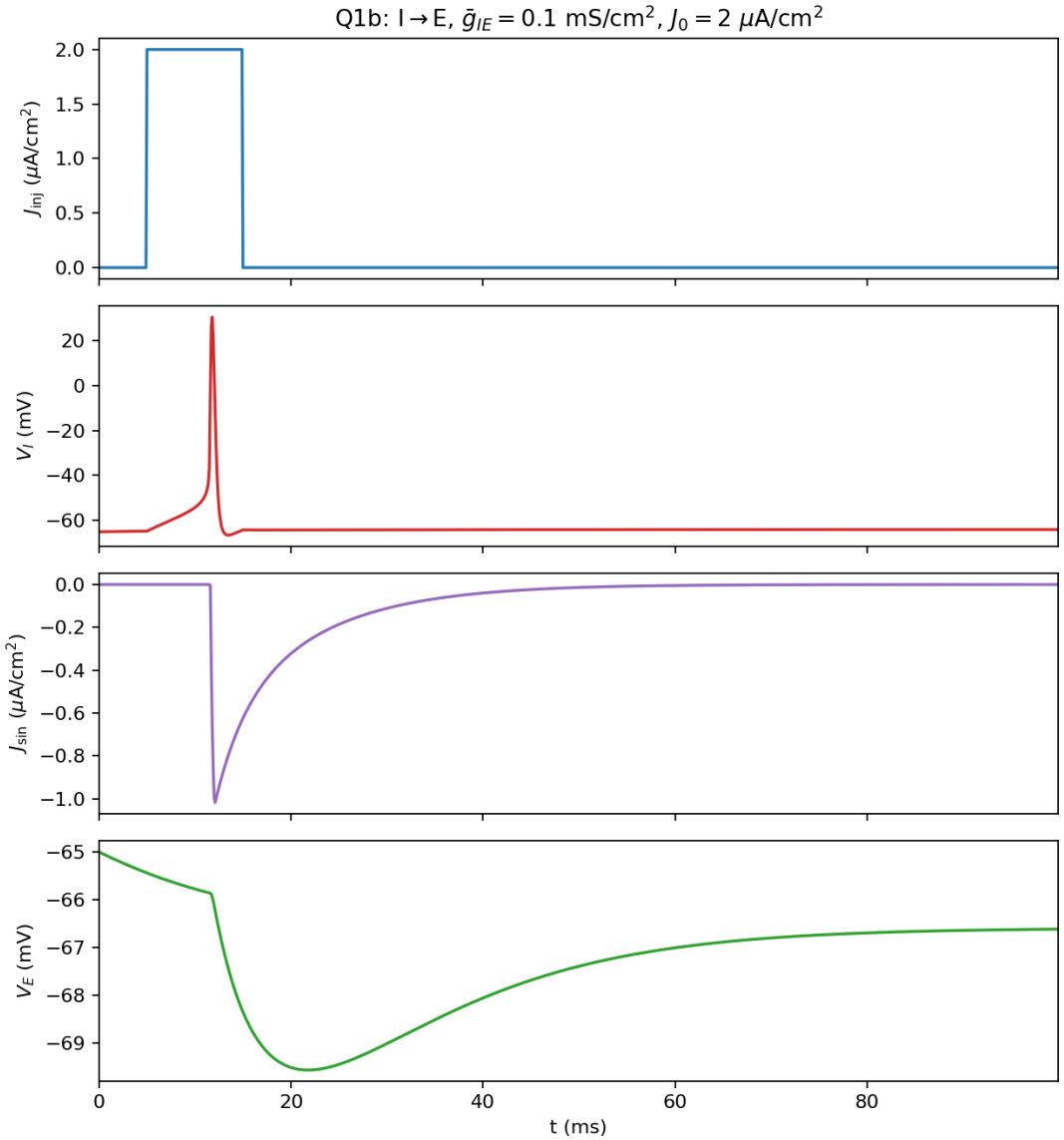


Figure 2: Question 1(b): I→E connection with $\bar{g}_{IE} = 0.1\,\mathrm{mS/cm^2}$ and $J_0 = 2\,\mu\mathrm{A/cm^2}$. From top to bottom: applied current, presynaptic voltage V_I, synaptic current J_{sin}, and postsynaptic voltage V_E.

The I neuron fires once, at $t = 11.6$ ms. Because $E_{\mathrm{rev}} = -80\,\mathrm{mV} < V_E$, the synaptic current is negative, with peak magnitude $\simeq 1.02\,\mu\mathrm{A/cm^2}$. The E neuron does not fire; its voltage, which relaxes towards a resting value of about -66.6 mV, is pushed down to a minimum of -69.6 mV at $t = 21.8$ ms (an inhibitory postsynaptic potential of roughly

3 mV) and then recovers. Within the 100 ms window no postsynaptic spike appears. The effect of the synapse is thus opposite in sign to that of item (a).

(c) Repetitive firing and the effect of $\bar{g}_{EI}$

The E neuron receives a current step of $J_0 = 1.5\,\mu\text{A/cm}^2$ from $t = 5$ ms to $t = 450$ ms (simulation of 500 ms) and drives the I neuron. The E neuron fires regularly, 26 spikes in total, with interspike interval

$$T = 17.66\,\text{ms} \qquad \Longrightarrow \qquad f_E = \frac{1}{T} \simeq 56.6\,\text{Hz}.$$

(The last interval is longer, 17.95 ms, because the current is switched off at 450 ms.) The period does not depend on $\bar{g}_{EI}$, since there is no feedback from I to E in this configuration.

The I neuron emits a spike train for *all ten* conductances $\bar{g}_{EI} = 0.10, 0.12, \ldots, 0.28\,\text{mS/cm}^2$ (Table 1). Supplementary runs below the requested range show that the response begins between 0.01 and 0.02 mS/cm^2 (no I spike at 0.01; 4, 9, 13, 16, 22 and 26 spikes at 0.02, 0.03, 0.04, 0.05, 0.06 and 0.08 mS/cm^2), so 0.10 mS/cm^2 already lies in the regime where each E spike evokes an I spike. Figure 3 shows the voltages.

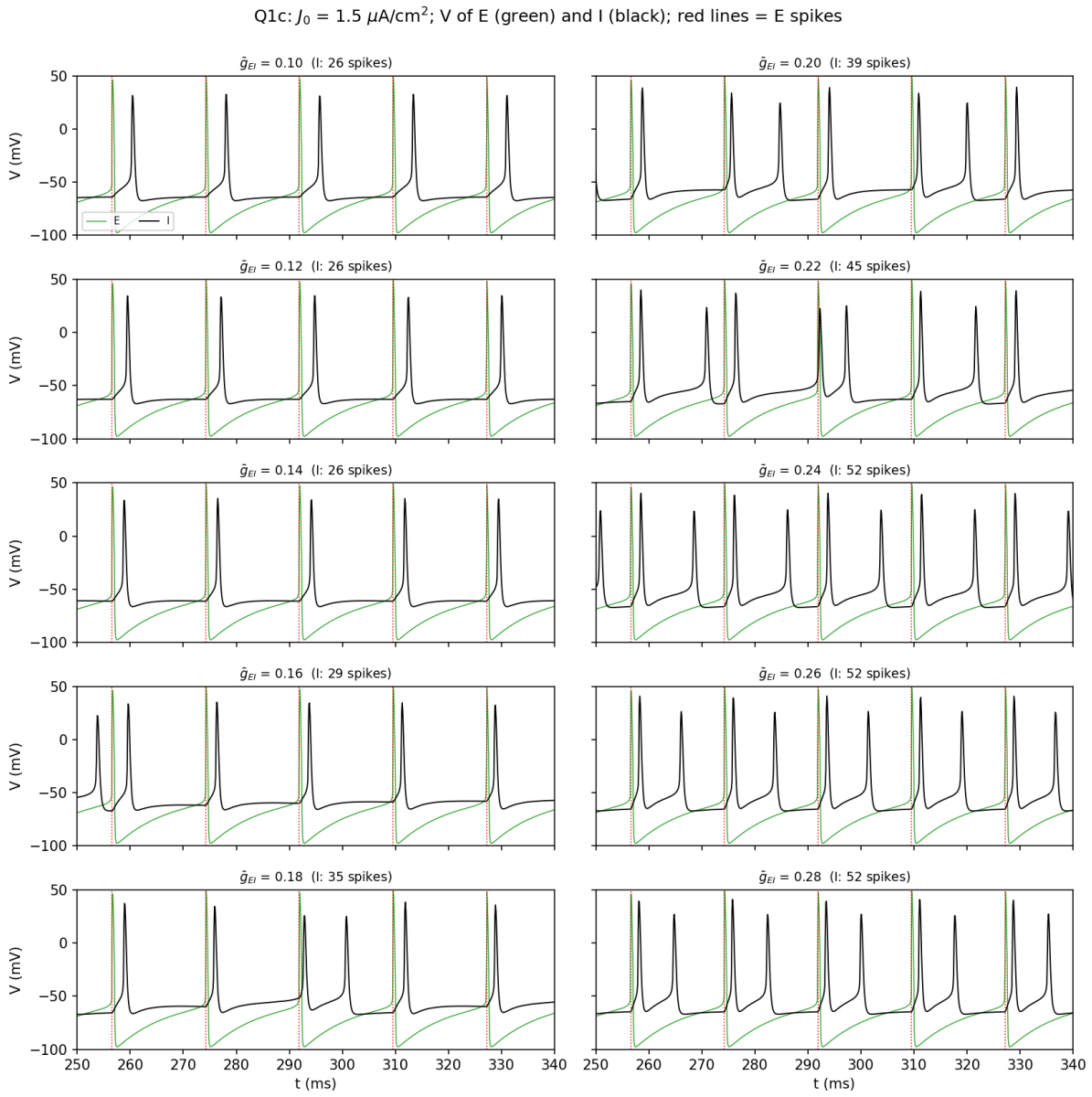


Figure 3: Question 1(c): voltages of the E (green) and I (black) neurons for the ten values of $\bar{g}_{EI}$, in a window of 90 ms taken during the steady firing. The red dotted lines mark the E spikes. For 0.10–0.14 mS/cm^2 the I neuron fires once per E spike; for larger $\bar{g}_{EI}$ extra I spikes appear between successive E spikes.

(d) Entrainment and phase locking

Entrainment means that the postsynaptic firing frequency equals (or is a simple multiple of) the presynaptic one; *phase locking* means that the delay δ between the presynaptic spike and the postsynaptic spike(s) is constant from cycle to cycle. For every E spike, δ was measured as the time from that spike to each I spike that follows it before the next E spike (Figure 4). A regime was classified as phase-locked only when the pattern of I spikes repeats every m E cycles with spike times matching within 0.15 ms over the last 14 repetitions of the window 100–445 ms. This criterion is deliberately strict, and the window contains only about 18 cycles, so a regime labelled "no strict locking" may be a slow transient or quasi-periodic rather than truly irregular.

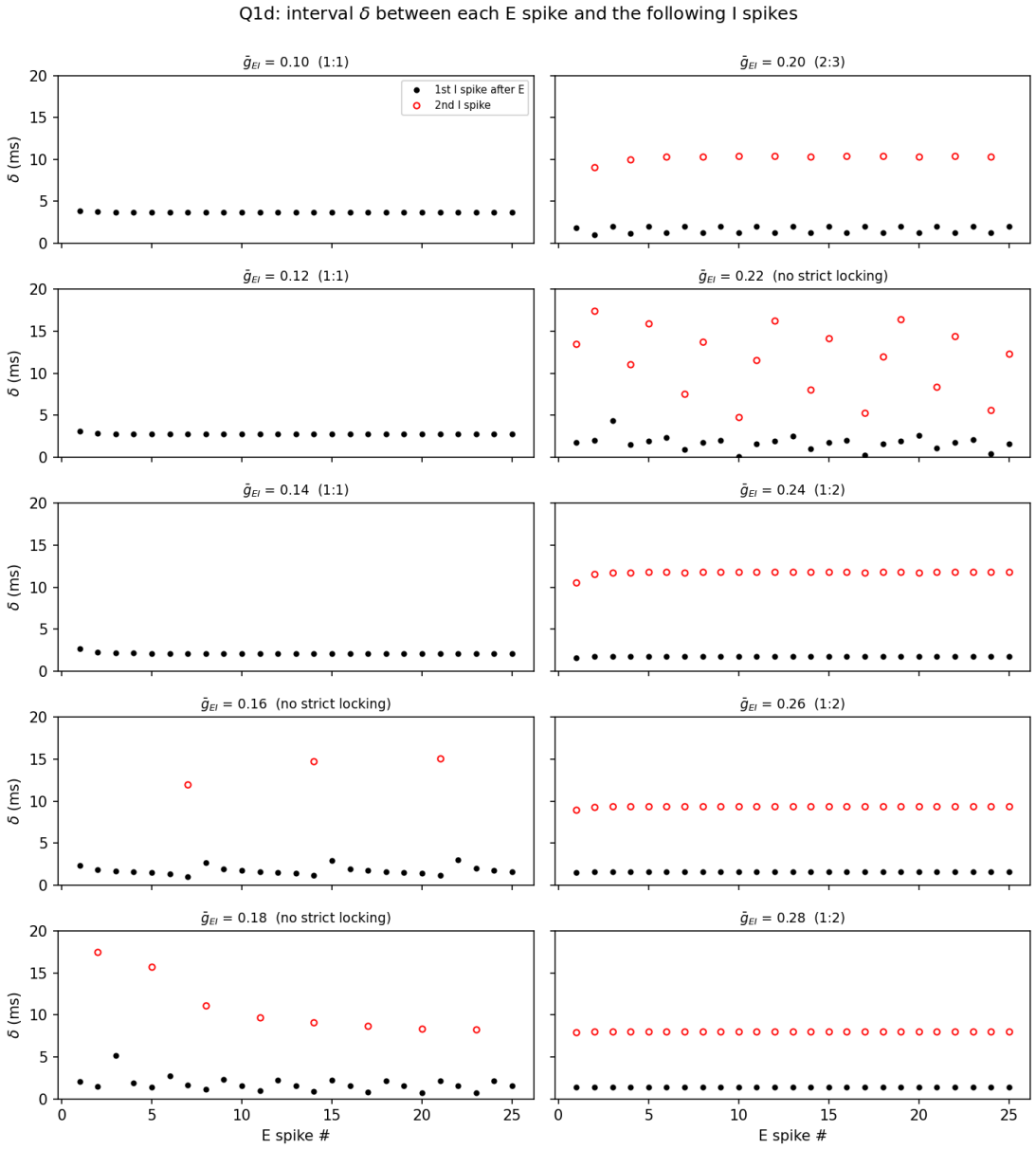


Figure 4: Question 1(d): interval δ between each E spike and the I spikes that follow it, for the ten values of $\bar{g}_{EI}$ (black: first I spike after the E spike; red circles: second I spike in the same cycle). Titles give the locking ratio (E spikes : I spikes).

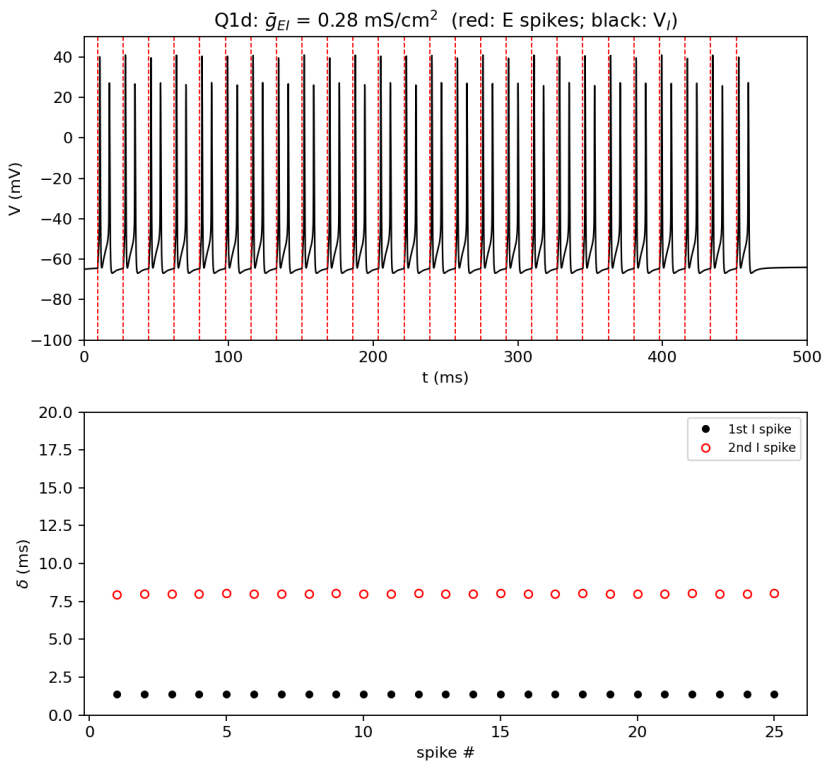


Figure 5: Question 1(d), example in the format of Figure 3 of the statement ($\bar{g}_{EI} = 0.28\,\mathrm{mS/cm^2}$). Top: V_I (black) with the E spike times (red dashed lines). Bottom: δ for each E spike (black: first I spike, red circles: second).

$\bar{g}_{EI}$ (mS/cm^2)	I spikes (E: 26)	E:I locking	δ (ms)
0.10	26	1:1	3.69
0.12	26	1:1	2.74
0.14	26	1:1	2.10
0.16	29	no strict locking	mostly one I spike per cycle with an occasional extra spike
0.18	35	no strict locking	extra I spike every third cycle, but its δ drifts from $\simeq 17$ to $\simeq 8$ ms
0.20	39	2:3	first spike alternates between $\simeq 1.2$ and $\simeq 2.0$; second spike at $\simeq 10.3$
0.22	45	no strict locking	irregular pattern of one or two I spikes per cycle
0.24	52	1:2	1.75 and 11.8
0.26	52	1:2	1.55 and 9.36
0.28	52	1:2	1.40 and 8.02

Table 1: Question 1(d): firing pattern of the I neuron for each $\bar{g}_{EI}$ ($J_0 = 1.5\,\mu\mathrm{A/cm^2}$).

Answer. Frequency entrainment with phase locking in the 1:1 sense (one postsynaptic spike per presynaptic spike, constant δ) occurs for $\bar{g}_{EI} = 0.10$, 0.12 and 0.14 mS/cm^2. In the notation of the statement, "n:1" entrainment ($n \geq 2$ presynaptic spikes per postsynaptic spike) was *not* observed for any $\bar{g}_{EI}$, because the I neuron never skips an E spike;

for larger $\bar{g}_{EI}$ it instead fires *more* often than E. In E:I terms, locked regimes with two I spikes per E spike (1:2, for 0.24, 0.26 and 0.28 mS/cm^2) and with three I spikes per two E spikes (2:3, for 0.20 mS/cm^2) were found. For $\bar{g}_{EI} = 0.16$, 0.18 and 0.22 mS/cm^2 no strict locking was detected in 450 ms. In the 1:2 cases the two intervals between I spikes sum to the E period (e.g. $11.04 + 6.62 = 17.66$ ms $= T$ for 0.28 mS/cm^2), as expected for locking.

Summary of Question 1

Item	Parameters	Result
(a)	E→I, $\bar{g}_{EI} = 0.05$, $J_0 = 3\,\mu$A/cm^2	E spike at 7.5 ms; I spike at 21.8 ms (latency 14.3 ms). Any J_0 from 0.6 to 3.8 μA/cm^2 (tested values) gives one E spike.
(b)	I→E, $\bar{g}_{IE} = 0.1$, $J_0 = 2\,\mu$A/cm^2	One I spike at 11.6 ms; E is inhibited ($V_E^{\min} = -69.6$ mV), no E spike. $J_0 = 4\,\mu$A/cm^2 gives two I spikes and does not qualify.
(c)	$J_0 = 1.5\,\mu$A/cm^2 step, ten values of $\bar{g}_{EI}$	$T = 17.66$ ms (56.6 Hz). The I neuron fires a train for all ten values.
(d)	same	1:1 locking for 0.10–0.14 mS/cm^2; 2:3 for 0.20; 1:2 for 0.24–0.28; no strict locking for 0.16, 0.18, 0.22. No n:1 with $n \geq 2$.

Table 2: Summary of Question 1.

Question 2: Rhythms generated by an inhibitory network (ING)

The network has five I and five O neurons: every O projects to every I ($\bar{g}_{OI} = 0.04$ mS/cm^2), every I projects to every O ($\bar{g}_{IO} = 0.12$ mS/cm^2) and every I projects to every I, including itself ($\bar{g}_{II} = 0.02$ mS/cm^2); there are no O–O connections. All I neurons receive $J = 1\,\mu$A/cm^2 and all O neurons $J = -3\,\mu$A/cm^2. The O neurons start from the values given in the statement, and the initial V of each I neuron is drawn uniformly from $[-70, -60]$ mV with h and n at their stationary values. The simulation lasts 1500 ms. Frequencies and the synchrony index below were computed for $t > 500$ ms. The index is $\chi = \sigma_{\bar{V}}/\langle\sigma_{V_i}\rangle_i$, the standard deviation (in time) of the population-mean voltage divided by the mean of the individual standard deviations; $\chi = 1$ means perfect synchrony.

(a) ING network with O→I synapses

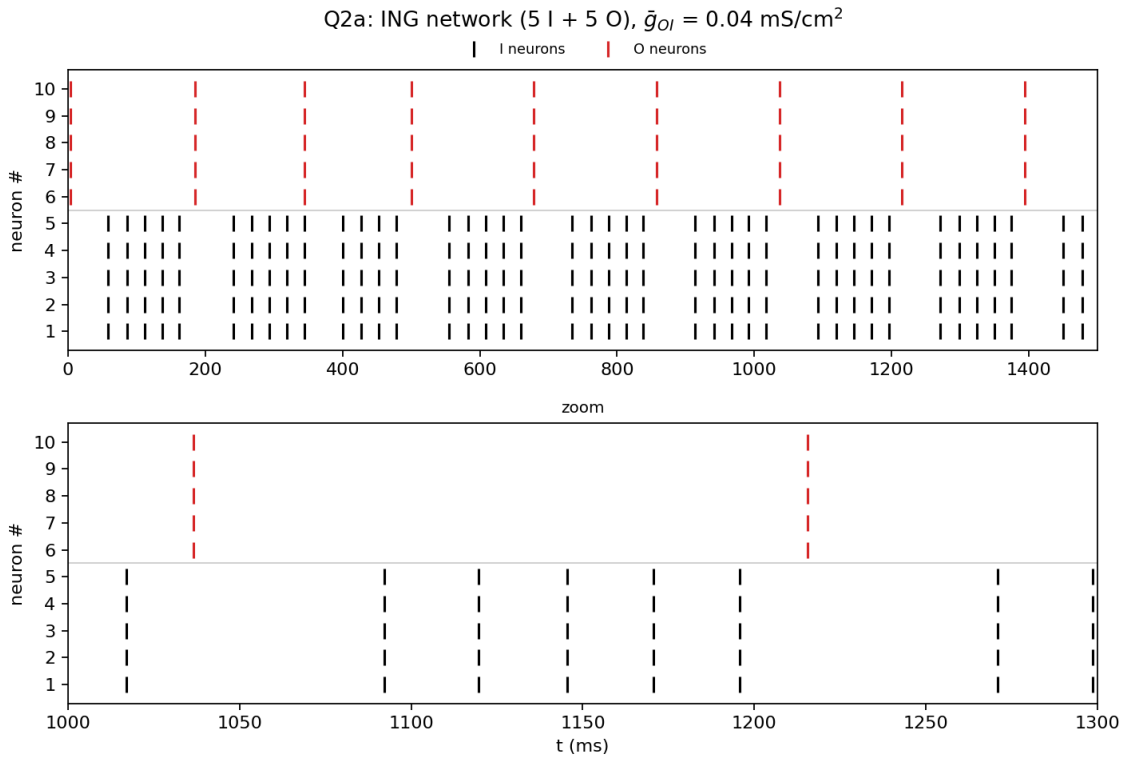


Figure 6: Question 2(a): raster plot of the network with $\bar{g}_{OI} = 0.04\,\mathrm{mS/cm^2}$ (neurons 1–5: I, black; neurons 6–10: O, red). Top: 1500 ms. Bottom: zoom on 1000–1300 ms.

The I neurons fire in bursts of gamma-frequency spikes (interspike interval $\simeq 25$ ms, i.e. $\simeq 39$ Hz), separated by silent gaps of $\simeq 75$ ms instead of 25 ms: an O volley arrives about 20 ms after the last I spike of a burst and postpones the next I spike (in the cycles inspected, for example at 500, 679 and 858 ms); their mean rate is 27 Hz. The O neurons fire in volleys at a rate of $\simeq 5.6$ Hz (interval 178.9 ms), in the theta band. The five I neurons converge to identical spike times (they are already coincident from the first I spike, at 58.6 ms) and the five O neurons are synchronous from the start ($\chi = 1.00$ for both populations). The O neurons are synchronous trivially, since they have identical initial conditions and inputs. The I neurons synchronize because they receive a common, strong inhibitory input from the first O volley at 2.9 ms and because of their mutual inhibition. So the network shows convergence to synchronous firing, with a gamma rhythm in I nested in a theta rhythm set by O. The O neurons fire repeatedly despite the negative injected current; this is presumably due to rebound from their h- and A-type currents, but that mechanism was not tested here.

(b) Without O→I synapses

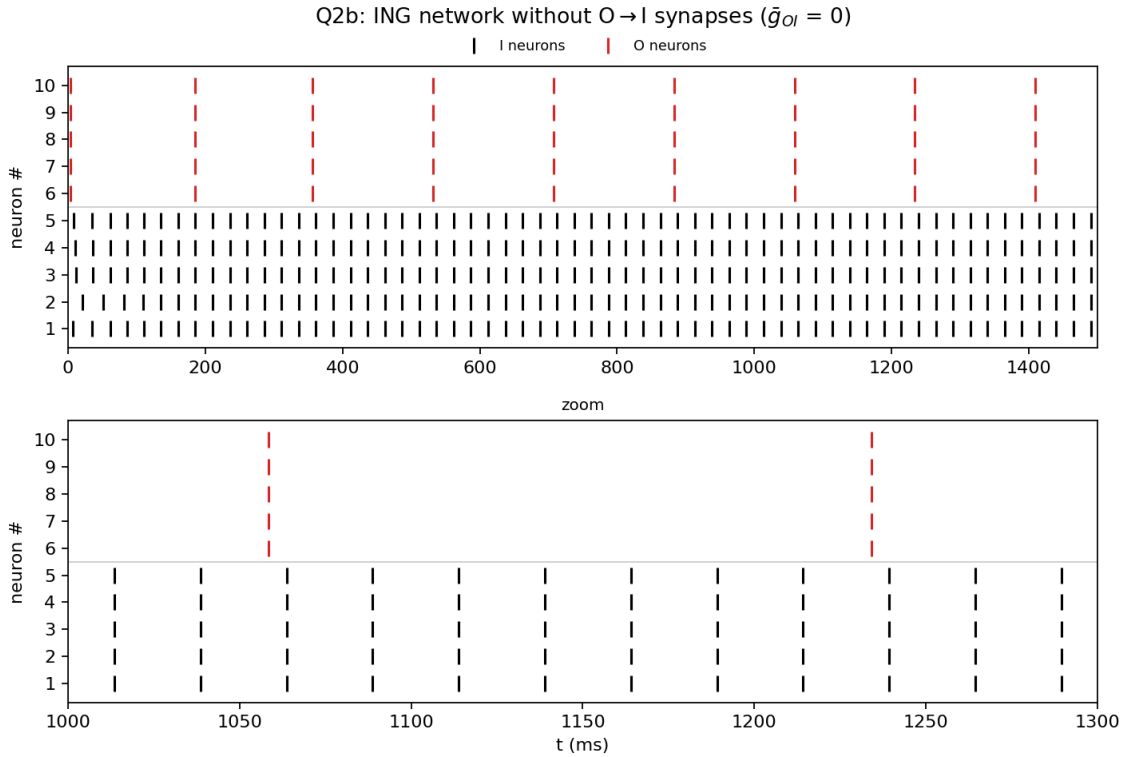


Figure 7: Question 2(b): raster plot with $\bar{g}_{OI} = 0$.

Without the O→I synapses the I population becomes a pure ING oscillator: all I neurons fire with a constant interval of 25.1 ms (39.9 Hz, mean rate 40 Hz), with no gaps (299 I spikes versus 205 in item (a)). The I neurons start with different phases (their first spikes lie between 7.5 and 20.6 ms) and converge to a synchronous rhythm ($\chi = 1.00$ for $t > 500$ ms). The O neurons still fire at the theta rate (5.7 Hz, interval 175.6 ms), now under an uninterrupted gamma-frequency inhibition from I. The main difference from (a) is thus the absence of the theta-timed pauses in the I population: the O→I synapses are what interrupts the gamma rhythm of I once per O cycle.

Summary of Question 2

Item	Condition	I neurons	O neurons
(a)	$\bar{g}_{OI} = 0.04$	gamma bursts (ISI $\simeq$ 25 ms, $\simeq$ 39 Hz) with $\simeq$ 75 ms pauses; mean rate 27 Hz; synchronous	5.6 Hz (theta); synchronous
(b)	$\bar{g}_{OI} = 0$	continuous gamma, ISI 25.1 ms (39.9 Hz); synchronous	5.7 Hz

Table 3: Summary of Question 2 ($t > 500$ ms).

Question 3: Rhythms in networks of excitatory and inhibitory neurons (PING)

(a) Minimal PING network

One E neuron and one I neuron are connected E→I and I→E (no autapses), with $\bar{g}_{EI} = \bar{g}_{IE} = 0.25\,\mathrm{mS/cm^2}$, $J_E = 1.4\,\mu\mathrm{A/cm^2}$ and $J_I = 0$. The statement does not give initial conditions for this item, so both neurons start at $V = -65\,\mathrm{mV}$ with h and n stationary. The simulation lasts 200 ms.

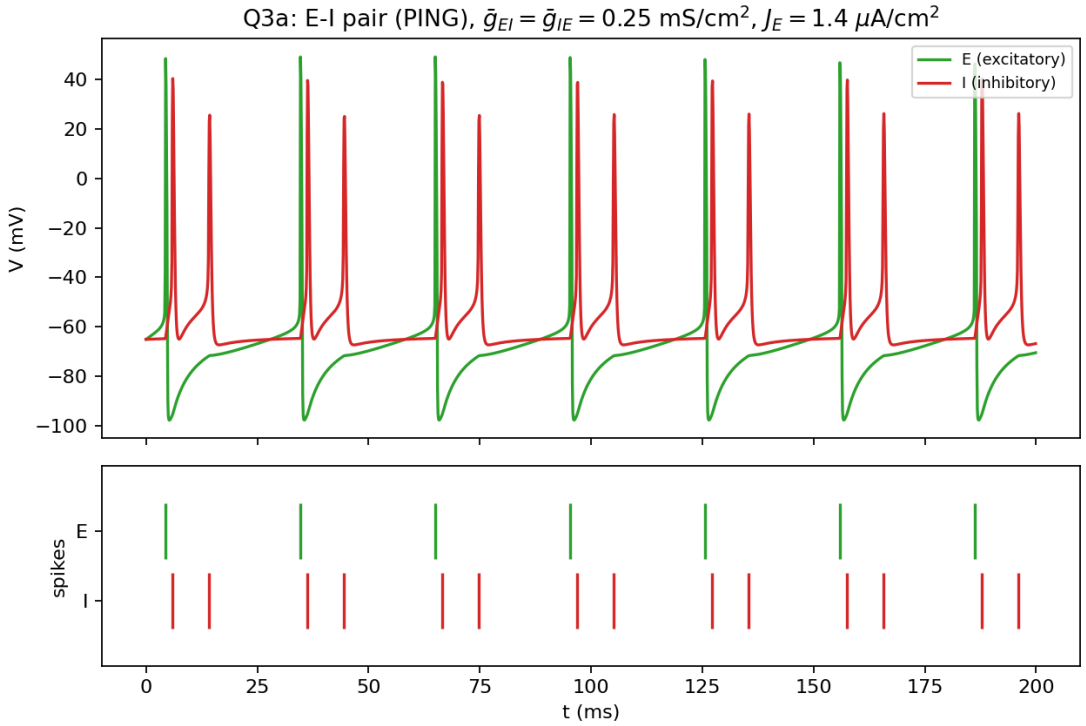


Figure 8: Question 3(a): voltages of the E (green) and I (red) neurons, with the corresponding spike times in the lower panel.

The two neurons oscillate periodically with $T = 30.3\,\mathrm{ms}$ (33 Hz, in the gamma band). In each cycle the E neuron fires first and the I neuron fires *twice*: the first I spike occurs $\simeq 1.5\,\mathrm{ms}$ after the E spike and the second one $\simeq 9.8\,\mathrm{ms}$ after it. The sequence is E, I, I, E, I, I, ... The mechanism is the one that gives PING its name: the E spike activates I through AMPA, I fires and the resulting $\mathrm{GABA_A}$ current silences E (note the hyperpolarization to $\simeq -97\,\mathrm{mV}$ after each E spike); E recovers when the inhibition decays and fires again. The double I spike is a property of this parameter set: the residual AMPA current from the E spike is strong enough ($\bar{g}_{EI} = 0.25\,\mathrm{mS/cm^2}$) to bring I back to threshold while E is still silenced.

(b) Sensitivity of the period

The period T was measured from the E spike times between $t = 300$ and 1000 ms (simulations of 1000 ms instead of 200 ms, to have enough cycles after the transient). The baseline is $T_0 = 30.310\,\mathrm{ms}$ (standard deviation of the intervals 0.008 ms), which agrees with the

value from the 200 ms run. Each parameter was perturbed by 1% and the sensitivity $S_p \simeq (\Delta T/T)/(\Delta p/p) \approx \partial \ln T / \partial \ln p$ was computed with a one-sided difference.

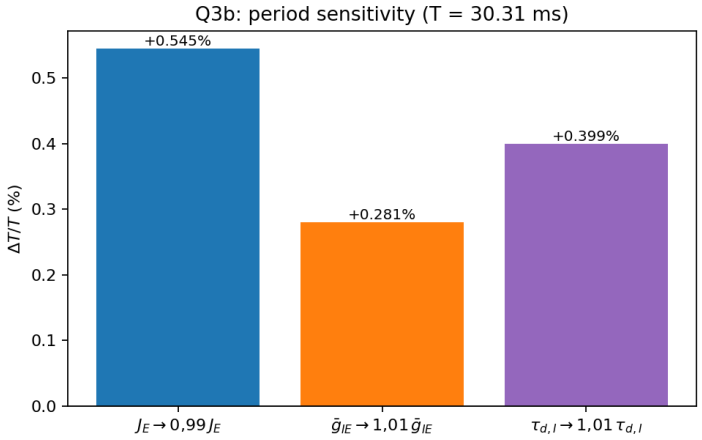


Figure 9: Question 3(b): percentage change of the period for each 1% perturbation.

Perturbation	T (ms)	$\Delta T/T$ (%)	S_p
baseline	30.310	—	—
$J_E \to 0.99\, J_E$	30.475	+0.545	−0.545
$\bar{g}_{IE} \to 1.01\, \bar{g}_{IE}$	30.395	+0.281	+0.281
$\tau_{d,I} \to 1.01\, \tau_{d,I}$	30.431	+0.399	+0.399

Table 4: Question 3(b): period of the E–I oscillation under 1% parameter changes.

The period is most sensitive to the current applied to the excitatory neuron, J_E: a 1% decrease lengthens T by 0.545%, which is about 1.4 times the effect of $\tau_{d,I}$ and about 1.9 times that of $\bar{g}_{IE}$. The signs are consistent with the mechanism: weaker drive to E, stronger inhibition or slower decay of the inhibition all delay the recovery of E and increase T. All changes are well above the cycle-to-cycle scatter of 0.008 ms in T, but they come from single-sided differences of 1% in one parameter around one operating point, so the ranking should not be extrapolated to large changes.

(c) E neuron added to the ING network

A single E neuron is added to the network of Question 2(a). It projects to all I and O neurons and receives synapses from all of them, with $\bar{g}_{IE} = 0.1$, $\bar{g}_{OE} = 0.15$, $\bar{g}_{EI} = 0.05$, $\bar{g}_{EO} = 0.01$, $\bar{g}_{OI} = 0.2$, $\bar{g}_{IO} = 0.5$ and $\bar{g}_{II} = 0.1\,\mathrm{mS/cm^2}$. The injected currents are $J_E = 0.8$, $J_I = 0.8$ and $J_O = -3\,\mu\mathrm{A/cm^2}$. The E neuron is initialized like the I neurons.

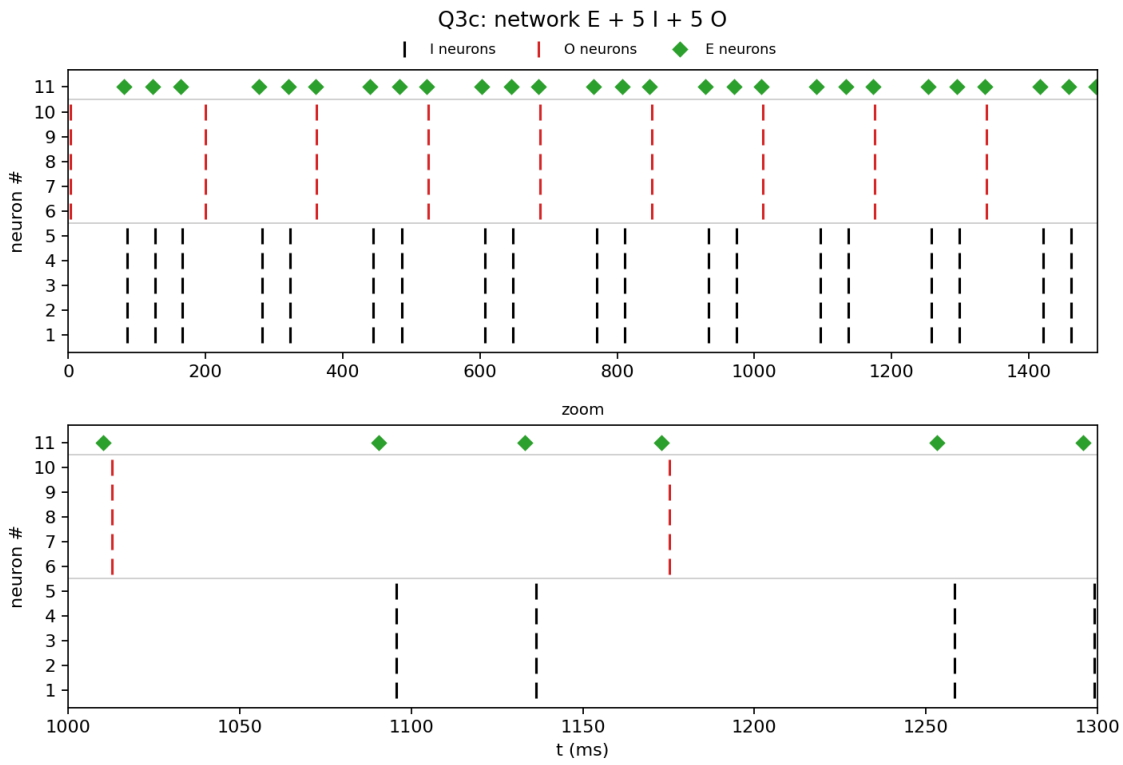


Figure 10: Question 3(c): raster plot of the network with one E neuron (green diamonds, top), five O neurons (red) and five I neurons (black).

The I neurons fire in short groups of two or three spikes with interspike interval $\simeq 41$ ms ($\simeq 24$ Hz), and the groups are separated by pauses of about 120–160 ms, so their mean rate over $t > 500$ ms is only 12 Hz (95 I spikes versus 205 in Question 2(a)). The E neuron fires in the same groups, with the E spike preceding the I spikes by $\simeq$ 3–5 ms (for example E at 807.8 ms and I at 811.1 ms; E at 928.0 ms and I at 933.0 ms), which is the PING order. The O volleys, at $\simeq 6$ Hz (interval 162.6 ms), fall inside the pauses of the I population. The E spike that occurs about 2.4 ms before an O volley (for example at 847.6 ms, with the O volley at 850.0 ms) is not followed by an I spike. The five I neurons are synchronous ($\chi = 1.00$).

Compared with the inhibitory-only network, the rhythm of the I population is therefore slower and sparser, and the gamma-range activity now comes with E leading I. This comparison must be read with care: besides adding E, the parameters $\bar{g}_{II}$, $\bar{g}_{OI}$, $\bar{g}_{IO}$ and J_I also differ from those of Question 2(a) (0.1 vs 0.02, 0.2 vs 0.04, 0.5 vs 0.12 mS/cm^2 and 0.8 vs 1 μA/cm^2), so the changes cannot be attributed to the E neuron alone; separating these effects would require additional simulations that were not run.

(d) Without O→I synapses

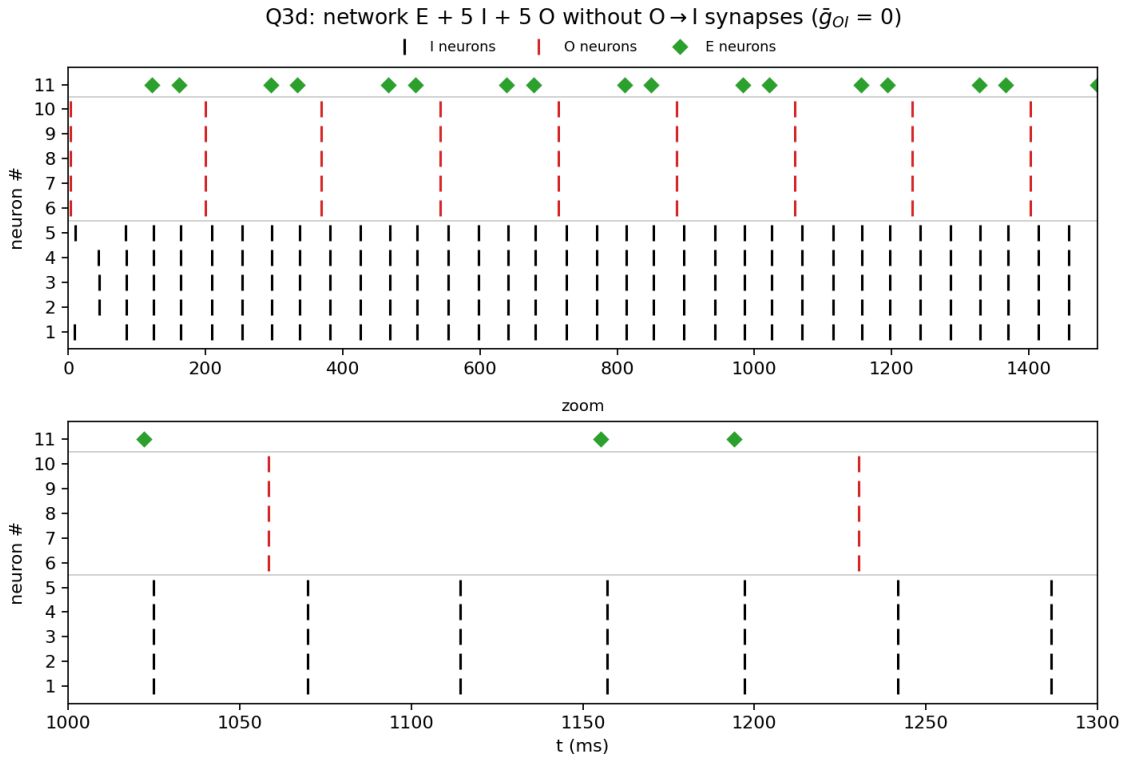


Figure 11: Question 3(d): raster plot of the network of item (c) with $\bar{g}_{OI} = 0$.

When the O→I synapses are removed, the I neurons are no longer silenced by the O volleys and fire almost continuously with an interval of $\simeq 44.6$ ms ($\simeq 22$ Hz; 170 I spikes against 95 in item (c)). The E neuron now fires in pairs of spikes ($\simeq 40$ ms apart), with one pair per O cycle: its pairs occur at 811, 983 and 1155 ms, separated by 172 ms, the period of the O neurons (5.8 Hz). The E neuron fires less often than in (c) (17 against 27 spikes in total). The I neurons remain synchronous ($\chi = 1.00$). In short, without O→I inhibition the E population fires one doublet per O cycle (it is locked to the O rhythm) instead of the I population being interrupted, as it does in item (c) and in Question 2(a).

Summary of Question 3

Item	Result
(a)	Order E, I, I per cycle; $T = 30.3$ ms (33 Hz). The first I spike follows E by $\simeq 1.5$ ms, the second by $\simeq 9.8$ ms.
(b)	$\Delta T/T$ for 1% perturbations: J_E (-1%): +0.545%; $\bar{g}_{IE}$ (+1%): +0.281%; $\tau_{d,I}$ (+1%): +0.399%. T is most sensitive to J_E.
(c)	Groups of 2–3 spikes of E and I ($\simeq 24$ Hz within a group), E leading I by 3–5 ms; O volleys at $\simeq 6$ Hz fall in the pauses; I mean rate 12 Hz.
(d)	I fires almost continuously ($\simeq 22$ Hz); E fires in doublets, one per O cycle (172 ms).

Table 5: Summary of Question 3.

Conclusions

The simulations illustrate three mechanisms by which synaptic coupling shapes neuronal activity. A single synapse changes the postsynaptic voltage with a sign set by $E_{\text{rev}} - V$, and sustained presynaptic firing can lock the postsynaptic neuron in regimes that depend on $\bar{g}$ (Question 1). A purely inhibitory network of I and O neurons synchronizes into a gamma rhythm in I, interrupted at the theta rate of O when the O→I synapses are present (Question 2). Adding an excitatory neuron produces the E-then-I ordering characteristic of PING, with a period that depends most strongly on the drive to E among the three parameters tested (Question 3). The statements about locking and about the networks of Question 3(c)–(d) are based on one realization of the random initial conditions and on windows of 450–1500 ms.

A Fortran code: `src/rede_EIO.f90`

```
! ======================================================================
! Lista 8 - Introducao a Neurociencia Computacional (FFCLRP-USP)
! Redes de neuronios E (piramidal, Olufsen), I (Wang-Buzsaki) e O (OLM,
! Tort et al.) com sinapses de Ermentrout-Kopell. Integracao: RK4, dt=0,01 ms.
!
! Uso:  ./rede_EIO caso [fJ] [fg] [ftau]
!   caso : q1a | q1b | q1c | q2a | q2b | q3a | q3b | q3c | q3d
!   (q1a,q1b,q1c: fJ = J0 [uA/cm2], fg = g sinaptico [mS/cm2])
!   fJ   : fator multiplicativo de J_E        (so q3b; padrao 1)
!   fg   : fator multiplicativo de g_IE       (so q3b; padrao 1)
!   ftau : fator multiplicativo de tau_d,I    (so q3b; padrao 1)
! Saidas (pasta ../data): <tag>_V.dat  (t, V_1..V_N a cada 0,1 ms)
!                         <tag>_spk.dat (indice, tipo, tempo do disparo)
! tipos: 1 = E, 2 = I, 3 = O
! ======================================================================
module rede
  implicit none
  integer, parameter :: dp = kind(1.0d0)
  integer, parameter :: NV = 8, NMAX = 12
  integer, parameter :: TE = 1, TI = 2, TO = 3
  integer :: N
  integer :: tipo(NMAX)
  real(dp) :: gmat(NMAX,NMAX)          ! gmat(i,j): pre i -> pos j  (mS/cm2)
  real(dp) :: Jinj(NMAX)               ! corrente injetada constante (uA/cm2)
  ! sinapses por tipo do neuronio PRE-sinaptico (AMPA, GABA_A(I), GABA_A(O))
  real(dp) :: taus(3) = (/ 0.1d0, 0.3d0, 0.2d0 /)
```

```
  real(dp) :: taud(3) = (/ 3.0d0, 9.0d0, 20.0d0 /)
  real(dp) :: Erev(3) = (/ 0.0d0, -80.0d0, -80.0d0 /)
  real(dp) :: Cm(3)   = (/ 1.0d0, 1.0d0, 1.3d0 /)
  ! pulso/degrau de corrente no neuronio jp (Q1): amplitude Jp0 em [tON,tOFF)
  integer  :: jp = 0
  real(dp) :: Jp0 = 0.0d0, tON = 0.0d0, tOFF = 0.0d0
  ! indices das variaveis: 1=V 2=h 3=n 4=m 5=a 6=b 7=r 8=s
contains

  ! x/(1-exp(-x/k)) com o limite x->0 tratado (singularidades removiveis)
  function vtrap(x, k) result(f)
    real(dp), intent(in) :: x, k
    real(dp) :: f
    if (abs(x/k) < 1.0d-7) then
      f = k
    else
      f = x/(1.0d0 - exp(-x/k))
    end if
  end function vtrap

  ! x_inf e tau_x das variaveis m, h, n de cada tipo (eqs. 4-12, 15-17)
  subroutine gates(tp, V, minf, taum, hinf, tauh, ninf, taun)
    integer, intent(in)  :: tp
    real(dp), intent(in)  :: V
    real(dp), intent(out) :: minf, taum, hinf, tauh, ninf, taun
    real(dp) :: am, bm, ah, bh, an, bn, fat
    select case (tp)
    case (TE)
      am = 0.32d0*vtrap(V+54.0d0, 4.0d0)
      bm = 0.28d0*vtrap(-(V+27.0d0), 5.0d0)
      ah = 0.128d0*exp(-(V+50.0d0)/18.0d0)
      bh = 4.0d0/(1.0d0 + exp(-(V+27.0d0)/5.0d0))
      an = 0.032d0*vtrap(V+52.0d0, 5.0d0)
      bn = 0.5d0*exp(-(V+57.0d0)/40.0d0)
      fat = 1.0d0
    case (TI)
      am = 0.1d0*vtrap(V+35.0d0, 10.0d0)
      bm = 4.0d0*exp(-(V+60.0d0)/18.0d0)
      ah = 0.07d0*exp(-(V+58.0d0)/20.0d0)
      bh = 1.0d0/(exp(-0.1d0*(V+28.0d0)) + 1.0d0)
      an = 0.01d0*vtrap(V+34.0d0, 10.0d0)
      bn = 0.125d0*exp(-(V+44.0d0)/80.0d0)
      fat = 0.2d0
    case default  ! TO
      am = 0.1d0*vtrap(V+38.0d0, 10.0d0)
      bm = 4.0d0*exp(-(V+65.0d0)/18.0d0)
      ah = 0.07d0*exp(-(V+63.0d0)/20.0d0)
      bh = 1.0d0/(1.0d0 + exp(-(V+33.0d0)/10.0d0))
      an = 0.018d0*vtrap(V-25.0d0, 25.0d0)
      bn = 0.0036d0*vtrap(-(V-35.0d0), 12.0d0)
      fat = 1.0d0
    end select
    minf = am/(am+bm);  taum = 1.0d0/(am+bm)
    hinf = ah/(ah+bh);  tauh = fat/(ah+bh)
    ninf = an/(an+bn);  taun = fat/(an+bn)
  end subroutine gates

  ! lado direito do sistema completo (rede de N neuronios)
  subroutine deriv(y, dy)
    real(dp), intent(in)  :: y(NV,NMAX)
    real(dp), intent(out) :: dy(NV,NMAX)
    integer  :: i, j
    real(dp) :: V, m, h, nn, Isyn, Iion
    real(dp) :: minf, taum, hinf, tauh, ninf, taun
    real(dp) :: ainf, binf, rinf, taub, taur
    dy = 0.0d0
    do j = 1, N
      V  = y(1,j);  h = y(2,j);  nn = y(3,j)
      Isyn = 0.0d0                                   ! eq. (23)
      do i = 1, N
        if (gmat(i,j) /= 0.0d0) Isyn = Isyn + gmat(i,j)*y(8,i)*(Erev(tipo(i)) - V)
      end do
      call gates(tipo(j), V, minf, taum, hinf, tauh, ninf, taun)
```

```
      select case (tipo(j))
      case (TE)
        m = minf                                          ! m = m_inf(V)
        Iion = 100.0d0*m**3*h*(50.0d0-V) + 80.0d0*nn**4*(-100.0d0-V) + 0.1d0*(-67.0d0-V)
      case (TI)
        m = minf
        Iion = 35.0d0*m**3*h*(55.0d0-V) + 9.0d0*nn**4*(-90.0d0-V) + 0.1d0*(-65.0d0-V)
      case default
        m = y(4,j)
        ainf = 1.0d0/(1.0d0 + exp(-(V+14.0d0)/16.6d0))
        binf = 1.0d0/(1.0d0 + exp((V+71.0d0)/7.3d0))
        taub = 1.0d0/(0.000009d0/exp((V-26.0d0)/18.5d0) + 0.014d0/(0.2d0+exp(-(V+70.0d0)
            /11.0d0)))
        rinf = 1.0d0/(1.0d0 + exp((V+84.0d0)/10.2d0))
        taur = 1.0d0/(exp(-14.59d0-0.086d0*V) + exp(-1.87d0+0.0701d0*V))
        Iion = 30.0d0*m**3*h*(90.0d0-V) + 23.0d0*nn**4*(-100.0d0-V) &
             + 16.0d0*y(5,j)*y(6,j)*(-90.0d0-V) + 12.0d0*y(7,j)*(-32.9d0-V) &
             + 0.05d0*(-70.0d0-V)
        dy(4,j) = (minf-m)/taum
        dy(5,j) = (ainf-y(5,j))/5.0d0
        dy(6,j) = (binf-y(6,j))/taub
        dy(7,j) = (rinf-y(7,j))/taur
      end select
      dy(1,j) = (Iion + Isyn + Jinj(j))/Cm(tipo(j))
      dy(2,j) = (hinf-h)/tauh
      dy(3,j) = (ninf-nn)/taun
      dy(8,j) = 0.5d0*(1.0d0+tanh(V/4.0d0))*(1.0d0-y(8,j))/taus(tipo(j)) &
              - y(8,j)/taud(tipo(j))                      ! eq. (22)
    end do
  end subroutine deriv

  ! condicao inicial "estacionaria" em h, n (e m, a, b, r) para um dado V
  subroutine ini_estac(j, V, y)
    integer, intent(in) :: j
    real(dp), intent(in) :: V
    real(dp), intent(inout) :: y(NV,NMAX)
    real(dp) :: minf, taum, hinf, tauh, ninf, taun
    call gates(tipo(j), V, minf, taum, hinf, tauh, ninf, taun)
    y(:,j) = 0.0d0
    y(1,j) = V;  y(2,j) = hinf;  y(3,j) = ninf;  y(4,j) = minf
    y(5,j) = 1.0d0/(1.0d0 + exp(-(V+14.0d0)/16.6d0))
    y(6,j) = 1.0d0/(1.0d0 + exp((V+71.0d0)/7.3d0))
    y(7,j) = 1.0d0/(1.0d0 + exp((V+84.0d0)/10.2d0))
  end subroutine ini_estac

  subroutine ini_0(j, y)          ! valores iniciais dados no enunciado (Q2a)
    integer, intent(in) :: j
    real(dp), intent(inout) :: y(NV,NMAX)
    y(:,j) = 0.0d0
    y(1,j) = -75.61d0;  y(4,j) = 0.0122d0;  y(3,j) = 0.07561d0
    y(2,j) = 0.9152d0;  y(7,j) = 0.06123d0; y(5,j) = 0.0229d0; y(6,j) = 0.2843d0
  end subroutine ini_0

  function sorteio() result(V)    ! V ~ U(-70,-60) mV
    real(dp) :: V, u
    call random_number(u)
    V = -70.0d0 + 10.0d0*u
  end function sorteio

end module rede

! ------------------------------------------------------------------------
program rede_EI0
  use rede
  implicit none
  character(len=16) :: caso, arg
  real(dp) :: fJ, fg, ftau
  real(dp) :: y(NV,NMAX), yn(NV,NMAX), k1(NV,NMAX), k2(NV,NMAX), k3(NV,NMAX), k4(NV,NMAX)
  real(dp), parameter :: dt = 0.01d0
  real(dp) :: tmax, t, tsp
  integer  :: nsteps, it, i, j, iE, nout, uV, uS, uQ
  integer, allocatable :: seed(:)
  integer :: nseed
```

```
172   character(len=64) :: fV, fS, fQ, tag
173
174   call get_command_argument(1, caso)
175   fJ = 1.0d0; fg = 1.0d0; ftau = 1.0d0
176   call get_command_argument(2, arg); if (len_trim(arg) > 0) read(arg,*) fJ
177   call get_command_argument(3, arg); if (len_trim(arg) > 0) read(arg,*) fg
178   call get_command_argument(4, arg); if (len_trim(arg) > 0) read(arg,*) ftau
179
180   ! semente fixa => resultados reprodutiveis
181   call random_seed(size=nseed); allocate(seed(nseed)); seed = 20261030
182   call random_seed(put=seed)
183
184   gmat = 0.0d0; Jinj = 0.0d0; y = 0.0d0; tipo = 0; iE = 0
185   select case (trim(caso))
186
187   case ('q2a', 'q2b')                       ! 5 I (1-5) + 5 O (6-10), mecanismo ING
188     N = 10; tmax = 1500.0d0
189     tipo(1:5) = TI; tipo(6:10) = TO
190     do i = 1, 5
191       do j = 1, 5
192         gmat(i,j) = 0.02d0                  ! I -> I (inclui autapse)
193       end do
194       do j = 6, 10
195         gmat(i,j) = 0.12d0                  ! I -> O
196         if (trim(caso) == 'q2a') gmat(j,i) = 0.04d0   ! O -> I (zero em q2b)
197       end do
198     end do
199     Jinj(1:5) = 1.0d0;  Jinj(6:10) = -3.0d0
200     do i = 1, 5; call ini_estac(i, sorteio(), y); end do
201     do i = 6, 10; call ini_O(i, y); end do
202
203   case ('q3a', 'q3b')                       ! par E-I, mecanismo PING
204     N = 2
205     if (trim(caso) == 'q3a') then; tmax = 200.0d0; else; tmax = 1000.0d0; end if
206     tipo(1) = TE; tipo(2) = TI
207     gmat(1,2) = 0.25d0                      ! E -> I
208     gmat(2,1) = 0.25d0*fg                   ! I -> E
209     taud(TI)  = 9.0d0*ftau
210     Jinj(1) = 1.4d0*fJ;  Jinj(2) = 0.0d0
211     call ini_estac(1, -65.0d0, y); call ini_estac(2, -65.0d0, y)
212
213   case ('q3c', 'q3d')                       ! 5 I (1-5) + 5 O (6-10) + 1 E (11)
214     N = 11; tmax = 1500.0d0; iE = 11
215     tipo(1:5) = TI; tipo(6:10) = TO; tipo(11) = TE
216     do i = 1, 5
217       do j = 1, 5
218         gmat(i,j) = 0.1d0                   ! g_II
219       end do
220       do j = 6, 10
221         gmat(i,j) = 0.5d0                   ! g_IO
222         if (trim(caso) == 'q3c') gmat(j,i) = 0.2d0    ! g_OI (zero em q3d)
223       end do
224       gmat(i,iE) = 0.1d0                    ! g_IE
225       gmat(iE,i) = 0.05d0                   ! g_EI
226     end do
227     do j = 6, 10
228       gmat(j,iE) = 0.15d0                   ! g_OE
229       gmat(iE,j) = 0.01d0                   ! g_EO
230     end do
231     Jinj(1:5) = 0.8d0;  Jinj(6:10) = -3.0d0;  Jinj(iE) = 0.8d0
232     do i = 1, 5; call ini_estac(i, sorteio(), y); end do
233     do i = 6, 10; call ini_O(i, y); end do
234     call ini_estac(iE, sorteio(), y)
235
236   case ('q1a', 'q1b', 'q1c')              ! Q1: par pre -> pos, corrente externa na pre
237     N = 2
238     if (trim(caso) == 'q1a') then; tmax = 30.0d0;  tOFF = 15.0d0; end if
239     if (trim(caso) == 'q1b') then; tmax = 100.0d0; tOFF = 15.0d0; end if
240     if (trim(caso) == 'q1c') then; tmax = 500.0d0; tOFF = 450.0d0; end if
241     tON = 5.0d0;  jp = 1;  Jp0 = fJ                   ! fJ = J0 (uA/cm2)
242     if (trim(caso) == 'q1b') then
243       tipo(1) = TI; tipo(2) = TE                      ! I -> E
244     else
```

```
      tipo(1) = TE; tipo(2) = TI                          ! E -> I
    end if
    gmat(1,2) = fg                                        ! fg = g_pre,pos (mS/cm2)
    call ini_estac(1, -65.0d0, y); call ini_estac(2, -65.0d0, y)

  case default
    write(*,*) 'caso invalido: ', trim(caso); stop 1
  end select

  tag = trim(caso)
  if (trim(caso) == 'q1c') write(tag,'(A,F4.2)') 'q1c_g', fg
  fV = '../data/' // trim(tag) // '_V.dat'
  fS = '../data/' // trim(tag) // '_spk.dat'
  fQ = '../data/' // trim(tag) // '_s.dat'
  if (trim(caso) == 'q3b') then
    if (fJ /= 1.0d0)   then; fV = '../data/q3b_J_V.dat';   fS = '../data/q3b_J_spk.dat';
        end if
    if (fg /= 1.0d0)   then; fV = '../data/q3b_g_V.dat';   fS = '../data/q3b_g_spk.dat';
        end if
    if (ftau /= 1.0d0) then; fV = '../data/q3b_tau_V.dat'; fS = '../data/q3b_tau_spk.dat'
       ; end if
  end if
  open(newunit=uV, file=trim(fV), status='replace')
  open(newunit=uS, file=trim(fS), status='replace')
  open(newunit=uQ, file=trim(fQ), status='replace')
  write(uV,'(A,*(1X,I1))') '# tipos', tipo(1:N)

  nsteps = nint(tmax/dt);  nout = 10
  do it = 0, nsteps - 1
    t = it*dt
    if (jp > 0) Jinj(jp) = merge(Jp0, 0.0d0, it >= nint(tON/dt) .and. it < nint(tOFF/dt))
    if (mod(it, nout) == 0) then
      write(uV,'(F10.3,*(1X,F9.4))') t, y(1,1:N)
      write(uQ,'(F10.3,*(1X,ES12.4))') t, y(8,1:N)
    end if
    call deriv(y, k1)
    yn = y + 0.5d0*dt*k1;  call deriv(yn, k2)
    yn = y + 0.5d0*dt*k2;  call deriv(yn, k3)
    yn = y + dt*k3;        call deriv(yn, k4)
    yn = y + dt/6.0d0*(k1 + 2.0d0*k2 + 2.0d0*k3 + k4)
    do j = 1, N                              ! disparo = cruzamento ascendente de 0 mV
      if (y(1,j) < 0.0d0 .and. yn(1,j) >= 0.0d0) then
        tsp = t + dt*(0.0d0 - y(1,j))/(yn(1,j) - y(1,j))
        write(uS,'(I3,1X,I2,1X,F12.5)') j, tipo(j), tsp
      end if
    end do
    y = yn
  end do
  close(uV); close(uS); close(uQ)
end program rede_EIO
```

B Shell script: `src/rodar_tudo.sh`

```
#!/bin/bash
# Lista 8 - compila, simula e gera as figuras das Questoes 2 e 3.
# Uso: cd src && ./rodar_tudo.sh
set -e
cd "$(dirname "$0")"
mkdir -p ../data ../figuras

echo ">> compilando"
gfortran -O2 -Wall -o rede_EIO rede_EIO.f90

echo ">> Questao 1 (par pre-pos)"
# varreduras auxiliares: numero de disparos do neuronio pre (col. 1 = indice 1) e pos (
    indice 2)
SCAN=../data/q1_varreduras.txt
echo "# caso  parametro  disparos_pre  disparos_pos" > $SCAN
for J in 0.2 0.4 0.6 0.8 1 1.5 2 3 3.5 3.8 4 5; do        # 1a: E->I, g_EI = 0,05
```

```
  ./rede_EIO q1a $J 0.05
  echo "q1a J0=$J $(awk '$1==1' ../data/q1a_spk.dat | wc -l) $(awk '$1==2' ../data/q1a_spk.dat | wc -l)" >> $SCAN
done
for J in 0.5 1 1.5 2 3 4; do                              # 1b: I->E, g_IE = 0,10
  ./rede_EIO q1b $J 0.10
  echo "q1b J0=$J $(awk '$1==1' ../data/q1b_spk.dat | wc -l) $(awk '$1==2' ../data/q1b_spk.dat | wc -l)" >> $SCAN
done
for g in 0.01 0.02 0.03 0.04 0.05 0.06 0.08; do           # inicio da resposta de I em 1c
  ./rede_EIO q1c 1.5 $g
  echo "q1c g=$g $(awk '$1==1' ../data/q1c_g${g}_spk.dat | wc -l) $(awk '$1==2' ../data/q1c_g${g}_spk.dat | wc -l)" >> $SCAN
  rm -f ../data/q1c_g${g}_*.dat
done
# rodadas finais (as que geram as figuras)
./rede_EIO q1a 3   0.05          # 1a: E->I, J0 = 3 uA/cm2, g_EI = 0,05
./rede_EIO q1b 2   0.10          # 1b: I->E, J0 = 2 uA/cm2, g_IE = 0,10
for g in 0.10 0.12 0.14 0.16 0.18 0.20 0.22 0.24 0.26 0.28; do
  ./rede_EIO q1c 1.5 $g          # 1c/1d: degrau de 1,5 uA/cm2 (5-450 ms), 500 ms
done

echo ">> Questao 2 (ING)"
./rede_EIO q2a          # 2a: 5 I + 5 O, gOI = 0,04
./rede_EIO q2b          # 2b: idem com gOI = 0

echo ">> Questao 3 (PING)"
./rede_EIO q3a                  # 3a: par E-I, 200 ms
./rede_EIO q3b                  # 3b: base (1000 ms, para medir T no regime estacionario)
./rede_EIO q3b 0.99 1    1      # (i)   J_E   -> 0,99 J_E
./rede_EIO q3b 1    1.01 1      # (ii)  g_IE  -> 1,01 g_IE
./rede_EIO q3b 1    1    1.01   # (iii) tau_d,I -> 1,01 tau_d,I
./rede_EIO q3c          # 3c: E + 5 I + 5 O
./rede_EIO q3d          # 3d: idem com gOI = 0

echo ">> figuras e metricas"
python3 graficos.py
echo ">> pronto: figuras em ../figuras, resumo em ../data/resumo.txt"
```

C Python script: `src/graficos.py`

```
#!/usr/bin/env python3
# -*- coding: utf-8 -*-
"""Lista 8 - Neurociencia Computacional: figuras e metricas das Questoes 2 e 3.
Le os arquivos de ../data gerados por rede_EIO.f90 e grava PNGs em ../figuras
e um resumo numerico em ../data/resumo.txt."""
import os
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

AQUI = os.path.dirname(os.path.abspath(__file__))
DATA = os.path.join(AQUI, "..", "data")
FIG = os.path.join(AQUI, "..", "figuras")
os.makedirs(FIG, exist_ok=True)
NOMES = {1: "E", 2: "I", 3: "O"}
COR = {1: "tab:green", 2: "k", 3: "tab:red"}
MARC = {1: "D", 2: "|", 3: "|"}
resumo = []

def log(s):
    print(s); resumo.append(s)

def le_V(tag):
    with open(os.path.join(DATA, tag + "_V.dat")) as f:
        tipos = np.array(f.readline().split()[2:], dtype=int)
    d = np.loadtxt(os.path.join(DATA, tag + "_V.dat"), skiprows=1)
    return d[:, 0], d[:, 1:], tipos
```

```
29
30  def le_spk(tag):
31      d = np.loadtxt(os.path.join(DATA, tag + "_spk.dat"), ndmin=2)
32      return d[:, 0].astype(int), d[:, 1].astype(int), d[:, 2]
33
34  def freq_pop(idx, tp, ts, tp_alvo, t0, t1, n_neur):
35      """taxa media de disparo (Hz) de uma populacao na janela [t0,t1]"""
36      m = (tp == tp_alvo) & (ts >= t0) & (ts <= t1)
37      return 1000.0 * m.sum() / (n_neur * (t1 - t0))
38
39  def isi_tipico(idx, tp, ts, tp_alvo, t0):
40      """ISI mediano (ms) dos disparos de um tipo apos t0 (por neuronio)"""
41      isis = []
42      for k in np.unique(idx[tp == tp_alvo]):
43          t = ts[(idx == k) & (ts >= t0)]
44          isis += list(np.diff(t))
45      return np.median(isis), np.mean(isis)
46
47  def chi_sync(t, V, tipos, tp_alvo, t0):
48      """indice de sincronia de Golomb-Rinzel (1 = sincronia perfeita)"""
49      m = t >= t0
50      X = V[m][:, tipos == tp_alvo]
51      if X.shape[1] < 2:
52          return np.nan
53      num = X.mean(axis=1).var()
54      den = X.var(axis=0).mean()
55      return np.sqrt(num/den)
56
57  def raster(tag, titulo, arq, zoom=(1000, 1300), t0_est=500):
58      t, V, tipos = le_V(tag)
59      idx, tp, ts = le_spk(tag)
60      N = len(tipos)
61      fig, ax = plt.subplots(2, 1, figsize=(9, 6.2))
62      for a, (x0, x1) in zip(ax, [(0, t[-1]), zoom]):
63          for tpa in (2, 3, 1):
64              m = (tp == tpa) & (ts >= x0) & (ts <= x1)
65              if m.any():
66                  a.plot(ts[m], idx[m], MARC[tpa], color=COR[tpa],
67                         ms=9 if tpa != 1 else 5, mew=1.4, ls="none",
68                         label=f"{NOMES[tpa]} neurons")
69          a.set_xlim(x0, x1); a.set_ylim(0.3, N + 0.7)
70          a.set_yticks(range(1, N + 1)); a.set_ylabel("neuron #")
71          for b in [5.5] + ([10.5] if N > 10 else []):
72              a.axhline(b, color="0.8", lw=0.8)
73      ax[0].set_title(titulo, pad=24)
74      ax[0].legend(loc="lower center", bbox_to_anchor=(0.5, 1.0), ncol=3, fontsize=8,
            frameon=False)
75      ax[1].set_xlabel("t (ms)")
76      ax[1].set_title("zoom", fontsize=9)
77      fig.tight_layout(); fig.savefig(os.path.join(FIG, arq), dpi=160); plt.close(fig)
78      # metricas
79      log(f"--- {tag} ({titulo}) ---")
80      for tpa in (2, 3, 1):
81          n = int((tipos == tpa).sum())
82          if n == 0: continue
83          med, mean = isi_tipico(idx, tp, ts, tpa, t0_est)
84          r = freq_pop(idx, tp, ts, tpa, t0_est, t[-1], n)
85          log(f"  tipo {NOMES[tpa]}: taxa media (t>{t0_est} ms) = {r:.1f} Hz | "
86              f"ISI mediano = {med:.1f} ms (f = {1000/med:.1f} Hz) | ISI medio = {mean:.1f}
                    ms (f = {1000/mean:.1f} Hz)"
87              f" | chi = {chi_sync(t, V, tipos, tpa, t0_est):.3f}")
88
89  # ------------------------------ Questao 2 ---------------------------------
90  raster("q2a", "Q2a: ING network (5 I + 5 O), $\\bar g_{OI}$ = 0.04 mS/cm$^2$", "
        Q2a_raster.png")
91  raster("q2b", "Q2b: ING network without O$\\rightarrow$I synapses ($\\bar g_{OI}$ = 0)",
        "Q2b_raster.png")
92
93  # ------------------------------ Questao 3a --------------------------------
94  t, V, tipos = le_V("q3a")
95  idx, tp, ts = le_spk("q3a")
96  fig, ax = plt.subplots(2, 1, figsize=(8, 5.5), sharex=True,
97                         gridspec_kw={"height_ratios": [2, 1]})
```

```
ax[0].plot(t, V[:, 0], color="tab:green", label="E (excitatory)")
ax[0].plot(t, V[:, 1], color="tab:red", label="I (inhibitory)")
ax[0].set_ylabel("V (mV)"); ax[0].set_title("Q3a: E-I pair (PING), $\\bar g_{EI}=\\bar g_{IE}=0.25$ mS/cm$^2$, $J_E=1.4\\ \\mu$A/cm$^2$")
ax[0].legend(loc="upper right", fontsize=8)
ax[1].eventplot([ts[idx == 1], ts[idx == 2]], colors=["tab:green", "tab:red"],
                lineoffsets=[1, 0], linelengths=0.8)
ax[1].set_yticks([0, 1]); ax[1].set_yticklabels(["I", "E"]); ax[1].set_xlabel("t (ms)")
ax[1].set_ylabel("spikes")
fig.tight_layout(); fig.savefig(os.path.join(FIG, "Q3a_VEI.png"), dpi=160); plt.close(fig)
tE = ts[idx == 1]; tI = ts[idx == 2]
log("--- q3a ---")
log(f"  disparos de E: {np.round(tE[:4],2)} ...; disparos de I: {np.round(tI[:5],2)} ...")
log(f"  T(E) medio em 200 ms = {np.diff(tE).mean():.2f} ms; I dispara {len(tI)/len(tE):.1f}x por ciclo de E")
log(f"  atraso do 1o disparo de I apos o de E (ciclo 2) = {tI[tI>tE[1]][0]-tE[1]:.2f} ms")

# ------------------------------ Questao 3b ---------------------------------
def periodo(tag, t_transiente=300.0):
    idx, tp, ts = le_spk(tag)
    t = ts[(idx == 1) & (ts > t_transiente)]
    return np.diff(t).mean(), np.diff(t).std(), len(t)

T0, s0, n0 = periodo("q3b")
log("--- q3b (simulacoes de 1000 ms, T medido com os disparos de E em t>300 ms) ---")
log(f"  base:        T = {T0:.4f} ms (desvio {s0:.1e}, {n0} disparos)  f = {1000/T0:.2f} Hz")
nomes = {"q3b_J": r"$J_E \to 0{,}99\,J_E$", "q3b_g": r"$\bar g_{IE} \to 1{,}01\,\bar g_{IE}$",
         "q3b_tau": r"$\tau_{d,I} \to 1{,}01\,\tau_{d,I}$"}
dT = {}
for k in nomes:
    T, s, n = periodo(k)
    dT[k] = 100 * (T - T0) / T0
    log(f"  {k:8s}: T = {T:.4f} ms  ->  variacao = {dT[k]:+.3f} %")
fig, ax = plt.subplots(figsize=(6.2, 4))
bars = ax.bar(range(3), [dT[k] for k in nomes], color=["tab:blue", "tab:orange", "tab:purple"])
ax.set_xticks(range(3)); ax.set_xticklabels([nomes[k] for k in nomes])
ax.axhline(0, color="k", lw=0.8)
for b, k in zip(bars, nomes):
    ax.text(b.get_x() + b.get_width()/2, dT[k], f"{dT[k]:+.3f}%", ha="center",
            va="bottom" if dT[k] >= 0 else "top", fontsize=9)
ax.set_ylabel("$\\Delta T / T$ (%)")
ax.set_title(f"Q3b: period sensitivity (T = {T0:.2f} ms)")
fig.tight_layout(); fig.savefig(os.path.join(FIG, "Q3b_sensibilidade.png"), dpi=160); plt.close(fig)
mais = max(dT, key=lambda k: abs(dT[k]))
log(f"  parametro com maior efeito sobre T: {mais}")

# ---------------------------- Questao 3c, 3d -------------------------------
raster("q3c", "Q3c: network E+5 I+5 0", "Q3c_raster.png")
raster("q3d", "Q3d: network E+5 I+5 0 without 0$\\rightarrow$I synapses ($\\bar g_{0I}$ = 0)", "Q3d_raster.png")


# ============================== QUESTAO 1 ====================================
# (rodar_tudo.sh gera: q1a [J0=3], q1b [J0=2], q1c_gX.XX [J0=1.5])
G_LISTA = ["0.10", "0.12", "0.14", "0.16", "0.18", "0.20", "0.22", "0.24", "0.26", "0.28"]

def le_q1(tag):
    d = np.loadtxt(os.path.join(DATA, tag + "_V.dat"), skiprows=1)
    s = np.loadtxt(os.path.join(DATA, tag + "_s.dat"))
    idx, tp, ts = le_spk(tag)
    return d[:, 0], d[:, 1], d[:, 2], s[:, 1], idx, ts

def fig_par(tag, J0, g, Erev, nome_pre, nome_pos, arq, titulo, cor_pre, cor_pos):
    t, Vpre, Vpos, s_pre, idx, ts = le_q1(tag)
    Jinj = np.where((t >= 5.0) & (t < 15.0), J0, 0.0)
```

```
    Jsin = g * s_pre * (Erev - Vpos)                        # eq. (21)
    fig, ax = plt.subplots(4, 1, figsize=(8, 8.5), sharex=True)
    ax[0].plot(t, Jinj, color="tab:blue"); ax[0].set_ylabel(r"$J_{\rm inj}$ ($\mu$A/cm$^2$)")
    ax[1].plot(t, Vpre, color=cor_pre); ax[1].set_ylabel(f"$V_{{{nome_pre}}}$ (mV)")
    ax[2].plot(t, Jsin, color="tab:purple"); ax[2].set_ylabel(r"$J_{\rm sin}$ ($\mu$A/cm$^2$)")
    ax[3].plot(t, Vpos, color=cor_pos); ax[3].set_ylabel(f"$V_{{{nome_pos}}}$ (mV)")
    ax[3].set_xlabel("t (ms)"); ax[0].set_title(titulo)
    for a in ax: a.set_xlim(0, t[-1])
    fig.tight_layout(); fig.savefig(os.path.join(FIG, arq), dpi=160); plt.close(fig)
    return t, Vpre, Vpos, Jsin, idx, ts

log("=== Questao 1 ===")
# --- (a) E -> I, J0 = 3, g_EI = 0,05
t, Vpre, Vpos, Jsin, idx, ts = fig_par("q1a", 3.0, 0.05, 0.0, "E", "I", "Q1a_E_to_I.png",
        r"Q1a: E$\rightarrow$I , $\bar g_{EI}=0.05$ mS/cm$^2$, $J_0=3\ \mu$A/cm$^2$", "tab:green", "tab:red")
log(f"  q1a: disparos de E = {np.round(ts[idx==1],2)}; disparos de I = {np.round(ts[idx==2],2)}; "
    f"max|Jsin| = {np.abs(Jsin).max():.3f}; V_I min = {Vpos.min():.2f} mV")
# --- (b) I -> E, J0 = 2, g_IE = 0,1
t, Vpre, Vpos, Jsin, idx, ts = fig_par("q1b", 2.0, 0.1, -80.0, "I", "E", "Q1b_I_to_E.png",
        r"Q1b: I$\rightarrow$E , $\bar g_{IE}=0.1$ mS/cm$^2$, $J_0=2\ \mu$A/cm$^2$", "tab:red", "tab:green")
log(f"  q1b: disparos de I (pre) = {np.round(ts[idx==1],2)}; disparos de E (pos) = {np.round(ts[idx==2],2)}; "
    f"V_E min = {Vpos.min():.2f} mV em t = {t[Vpos.argmin()]:.1f} ms; max|Jsin| = {np.abs(Jsin).max():.3f}")

# --- (c) degrau de 1,5 uA/cm2, varredura em g_EI
def ciclos(tE, tI, t0, t1):
    out = []
    for a, b in zip(tE[:-1], tE[1:]):
        if a < t0 or b > t1: continue
        out.append(tI[(tI >= a) & (tI < b)] - a)
    return out

def classifica(tE, tI):
    cyc = ciclos(tE, tI, 100.0, 445.0)
    n = len(cyc)
    for m in range(1, 13):
        if all(len(cyc[k]) == len(cyc[k+m]) and np.all(np.abs(cyc[k] - cyc[k+m]) < 0.15)
               for k in range(n-m-14, n-m)):
            q = sum(len(c) for c in cyc[-m:])
            return m, q
    return None, None

dados = {}
for g in G_LISTA:
    tag = f"q1c_g{g}"
    t, VE, VI, sE, idx, ts = le_q1(tag)
    tE = ts[idx == 1]; tI = ts[idx == 2]
    m, q = classifica(tE, tI)
    dados[g] = (t, VE, VI, tE, tI, m, q)
    TE = np.diff(tE[tE < 445]).mean()
    log(f"  q1c g={g}: nE={len(tE)} nI={len(tI)}  T_E={TE:.3f} ms ({1000/TE:.1f} Hz)  "
        + (f"travamento: {m} disparo(s) de E : {q} de I" if m else "sem travamento estrito em 450 ms"))

fig, ax = plt.subplots(5, 2, figsize=(11, 11), sharex=True, sharey=True)
for a, g in zip(ax.T.ravel(), G_LISTA):
    t, VE, VI, tE, tI, m, q = dados[g]
    a.plot(t, VE, color="tab:green", lw=0.7, label="E")
    a.plot(t, VI, color="k", lw=0.9, label="I")
    for x in tE: a.axvline(x, color="r", ls=":", lw=0.8)
    a.set_xlim(250, 340); a.set_ylim(-100, 50)
    a.set_title(rf"$\bar g_{{EI}}$ = {g}  (I: {len(tI)} spikes)", fontsize=9)
for a in ax[-1]: a.set_xlabel("t (ms)")
for a in ax[:, 0]: a.set_ylabel("V (mV)")
ax[0, 0].legend(loc="lower left", fontsize=7, ncol=2)
```

```
fig.suptitle("Q1c: $J_0$ = 1.5 $\\mu$A/cm$^2$; V of E (green) and I (black); red lines = E spikes", y=0.995)
fig.tight_layout(); fig.savefig(os.path.join(FIG, "Q1c_sweep.png"), dpi=150); plt.close(fig)

# --- (d) sincronizacao e travamento de fase: delta = I - E
def deltas(tE, tI):
    d1, d2 = [], []
    for a, b in zip(tE[:-1], tE[1:]):
        c = tI[(tI >= a) & (tI < b)] - a
        d1.append(c[0] if len(c) > 0 else np.nan)
        d2.append(c[1] if len(c) > 1 else np.nan)
    return np.array(d1), np.array(d2)

fig, ax = plt.subplots(5, 2, figsize=(10, 11), sharex=True, sharey=True)
for a, g in zip(ax.T.ravel(), G_LISTA):
    t, VE, VI, tE, tI, m, q = dados[g]
    d1, d2 = deltas(tE, tI)
    k = np.arange(1, len(d1) + 1)
    a.plot(k, d1, "ko", ms=3, label="1st I spike after E")
    a.plot(k, d2, "o", mfc="none", mec="r", ms=4, label="2nd I spike")
    rot = f"{m}:{q}" if m else "no strict locking"
    a.set_title(rf"$\bar g_{{EI}}$ = {g}  ({rot})", fontsize=9)
    a.set_ylim(0, 20)
for a in ax[-1]: a.set_xlabel("E spike #")
for a in ax[:, 0]: a.set_ylabel(r"$\delta$ (ms)")
ax[0, 0].legend(loc="upper right", fontsize=7)
fig.suptitle(r"Q1d: interval $\delta$ between each E spike and the following I spikes", y=0.995)
fig.tight_layout(); fig.savefig(os.path.join(FIG, "Q1d_delta.png"), dpi=150); plt.close(fig)

# exemplo no formato da Figura 3 do enunciado: g = 0,28
t, VE, VI, tE, tI, m, q = dados["0.28"]
fig, ax = plt.subplots(2, 1, figsize=(7.5, 7))
ax[0].plot(t, VI, "k", lw=1)
for x in tE: ax[0].axvline(x, color="r", ls="--", lw=0.8)
ax[0].set_xlim(0, 500); ax[0].set_ylim(-100, 50)
ax[0].set_xlabel("t (ms)"); ax[0].set_ylabel("V (mV)")
ax[0].set_title(r"Q1d: $\bar g_{EI}$ = 0.28 mS/cm$^2$  (red: E spikes; black: V$_I$)")
d1, d2 = deltas(tE, tI); k = np.arange(1, len(d1) + 1)
ax[1].plot(k, d1, "ko", ms=4, label="1st I spike"); ax[1].plot(k, d2, "o", mfc="none", mec="r", ms=5, label="2nd I spike")
ax[1].set_xlabel("spike #"); ax[1].set_ylabel(r"$\delta$ (ms)"); ax[1].set_ylim(0, 20); ax[1].legend(fontsize=8)
fig.tight_layout(); fig.savefig(os.path.join(FIG, "Q1d_g_0.28.png"), dpi=160); plt.close(fig)

with open(os.path.join(DATA, "resumo.txt"), "w") as f:
    f.write("\n".join(resumo) + "\n")
```